

VFSTR - VADLAMUDI Vadlamudi	Regd. No.									
	Year	Semester			Branch			Section		
	I	II			CSE			1 - 13		
	Course:	EP								
Program Name: B. Tech.	Date: 10.03.26	Time: 11.00 to 12.30 PM (Each Course 30 Minutes)					Pre-TI			

Time allotted for each course assessment: 30 Minutes.

Max. Marks: 10 Marks

- ❖ Answer all two question from each course
- ❖ Answers should be written for course wise in different sheets

EP (25PY101)

1. Elucidate the concept of doping and explain the formation of n-type and p-type semiconductors? **5M**

2. a) Write the expression for the de-Broglie wavelength in the following cases **3M**
 - i) A particle is moving with a kinetic energy 'E'.
 - ii) A particle is placed between the two electrodes having a potential difference of 'V' volts.
 - iii) A particle is at a thermal equilibrium of temperature 'T'.

- b) i) Calculate the de-Broglie's wavelength of an electron moving with a kinetic energy of 1MeV.
- ii) Calculate the de-Broglie wavelength of an electron when a potential difference of 6400V is applied. **2M**